Lab: Motor Control -- How Your Brain Moves Your Body

Objective

Investigate how the nervous system controls **voluntary and reflexive action** through motor predictions, and explore how Active Inference explains motor control as fulfilling proprioceptive predictions.

Prerequisites

- Completed Biology modules on Systems through Cognition
- Read the module on Motor Control, Reflexes, and Voluntary Action

Part 1: Reflex Arc Mapping

Goal: Map the reflex arc as the fastest biological action loop.

Research the knee-jerk (patellar) reflex and map each step:

Step	Structure Involved	What Happens	Active Inference Interpretation
1. Stimulus			Sensory perturbation
2. Sensory neuron			Prediction error signal
3. Spinal cord integration			Rapid model comparison
4. Motor neuron			Action command
5. Muscle response			Prediction fulfillment

Why are reflexes faster than voluntary actions? What does this tell us about where in the hierarchy the "model" lives?

Write your response here

Part 2: Proprioception Experiment

Goal: Experience proprioception (your sense of body position) as a prediction system.

Experiment: Close your eyes and slowly touch your nose with your index finger. Try it 5 times with each hand.

Trial	Hand	Success?	How Far Off?	What Happened?
1	Right			
2	Right			
3	Left			
4	Left			
5	Both simultaneously			

In Active Inference, voluntary movement works by the brain predicting where the limb *should* be. The motor system then moves the limb to fulfill that prediction. How does this experiment demonstrate that mechanism?

Write your response here

Part 3: Motor Learning -- Tracing with a Mirror

Goal: Observe motor learning as model updating in real time.

Draw a star shape while looking only at your hand's reflection in a mirror (not directly at your hand). Try it three times.

Attempt	Time to Complete	Number of Errors	How It Felt
1			
2			
3			

1. Why was the first attempt so difficult? (Your motor model predicts hand movement in one direction, but the mirror reverses it.)
2. Did you improve? That improvement is your cerebellum updating its forward model.
3. How does this relate to the Active Inference distinction between perception (updating beliefs) and action (moving to fulfill predictions)?

Write your response here

Part 4: Movement Disorders as Action Failures

Goal: Analyze a motor condition through the Active Inference lens.

Choose one: Parkinson's disease, cerebellar ataxia, or motor tics. Research and answer:

1. What motor predictions are disrupted?
2. What does the biological damage involve?
3. How do treatments attempt to restore proper motor prediction?

Write your response here

Summary Table

Concept	Biological Mechanism	Active Inference Connection
Reflex arc	Spinal cord loop	Fast, low-level prediction fulfillment
Proprioception	Muscle spindles + cerebellum	Predicting body position
Motor learning	Cerebellar forward model	Updating action predictions through practice
Movement disorders	Basal ganglia / cerebellum damage	Disrupted motor predictions